

2.1.4 Acids

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Acids, bases, alkalis and neutralisation	
(a) the formulae of the common acids (HCl , H_2SO_4 , HNO_3 and CH_3COOH) and the common alkalis (NaOH , KOH and NH_3) and explanation that acids release H^+ ions in aqueous solution and alkalis release OH^- ions in aqueous solution	
(b) qualitative explanation of strong and weak acids in terms of relative dissociations	
(c) neutralisation as the reaction of: (i) H^+ and OH^- to form H_2O (ii) acids with bases, including carbonates, metal oxides and alkalis (water-soluble bases), to form salts, including full equations	
Acid–base titrations	
(d) the techniques and procedures used when preparing a standard solution of required concentration and carrying out acid–base titrations	PAG2 HSW4 Many opportunities to carry out experimental and investigative work.
(e) structured and non-structured titration calculations, based on experimental results of familiar and non-familiar acids and bases.	<i>M0.1, M0.2, M1.1, M1.2, M2.2, M2.3, M2.4</i>